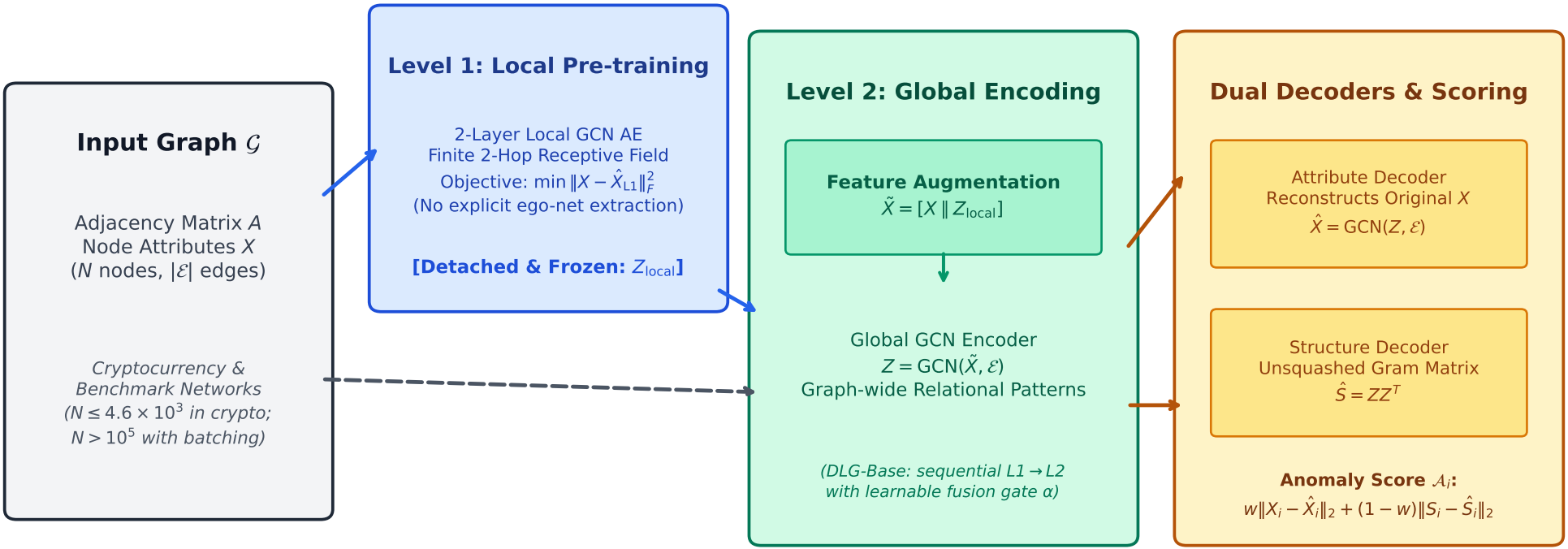

Input Graph $\mathcal{G}$
Adjacency Matrix A
Node Attributes X
(N nodes, $|\mathcal{E}|$ edges)
Cryptocurrency & Benchmark Networks
($N \leq 4.6 \times 10^3$ in crypto; $N > 10^5$ with batching)
Level 1: Local Pre-training
2-Layer Local GCN AE
Finite 2-Hop Receptive Field
Objective: $\min \|X - \hat{X}_{L1}\|_F^2$
(No explicit ego-net extraction)
[Detached & Frozen: Z_{local}]
Level 2: Global Encoding
Feature Augmentation
$\tilde{X} = [X \,\|\, Z_{local}]$
Global GCN Encoder
$Z = \text{GCN}(\tilde{X}, \mathcal{E})$
Graph-wide Relational Patterns
(DLG-Base: sequential L1 → L2 with learnable fusion gate α)
Dual Decoders & Scoring
Attribute Decoder
Reconstructs Original X
$\hat{X} = \text{GCN}(Z, \mathcal{E})$
Structure Decoder
Unsquashed Gram Matrix
$\hat{S} = ZZ^T$
Anomaly Score $\mathcal{A}_i$:
$w\|X_i - \hat{X}_i\|_2 + (1-w)\|S_i - \hat{S}_i\|_2$